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Project

Clean Up

```
clc
clear all
close all
```

A:Create a DH table for (PPP-RRR)

```
%theta, d, a, alpha, joint type
  L1 = Link([0 0 0 0 1]); %Prismatic
  L2 = Link([0 0 0 -pi/4 1]); %Prismatic
  L3 = Link([0 0 7 pi/3 1]); %Prismatic
  L4 = Link([0 0 0 pi/3 0]); %Revolute
  L5 = Link([0 7 0 -pi/2 0]); %Revolute
  L6 = Link([0 0 0 0 0]); %Revolute
  L1.qlim = [0 9];
  L2.qlim = [0 9];
  L3.qlim = [0 9];
  r = SerialLink([L1 L2 L3 L4 L5 L6])
```

```
r =

noname:: 6 axis, PPPRRR, stdDH, fastRNE
+---+-----------+-----------+-----------+-----------+-----------+
| j |     theta |         d |         a |     alpha |    offset |
+---+-----------+-----------+-----------+-----------+-----------+
|  1|         0|         q1|          0|          0|          0|
|  2|         0|         q2|          0|  -0.785398|          0|
|  3|         0|         q3|          7|     1.0472|          0|
|  4|        q4|          0|          0|     1.0472|          0|
|  5|        q5|          7|          0|    -1.5708|          0|
|  6|        q6|          0|          0|          0|          0|
+---+-----------+-----------+-----------+-----------+-----------+
```

B:Find the forward kinematics matrix

```
rob = fkine(r, [ 3 6 2  -pi/3  pi/4  pi/2])
```

```
rob =
    0.7500   -0.6597   -0.0474      1.75
    0.5477    0.5792    0.6038    -2.419
   -0.3709   -0.4788    0.7958     13.01
         0         0         0         1
```

C:Find the position (x, y, z) of the end-effector using the first three joints(assume the values)

```
ro = fkine(r, [ 3 6 2  0 0 0])
x = ro.t(1,1)
y = ro.t(2,1)
Z = ro.t(3,1)
```

```
ro =
         1         0         0         7
         0    0.9659    0.2588    -5.347
         0   -0.2588    0.9659     12.23
         0         0         0         1

x =

     7


y =

   -5.3473


Z =

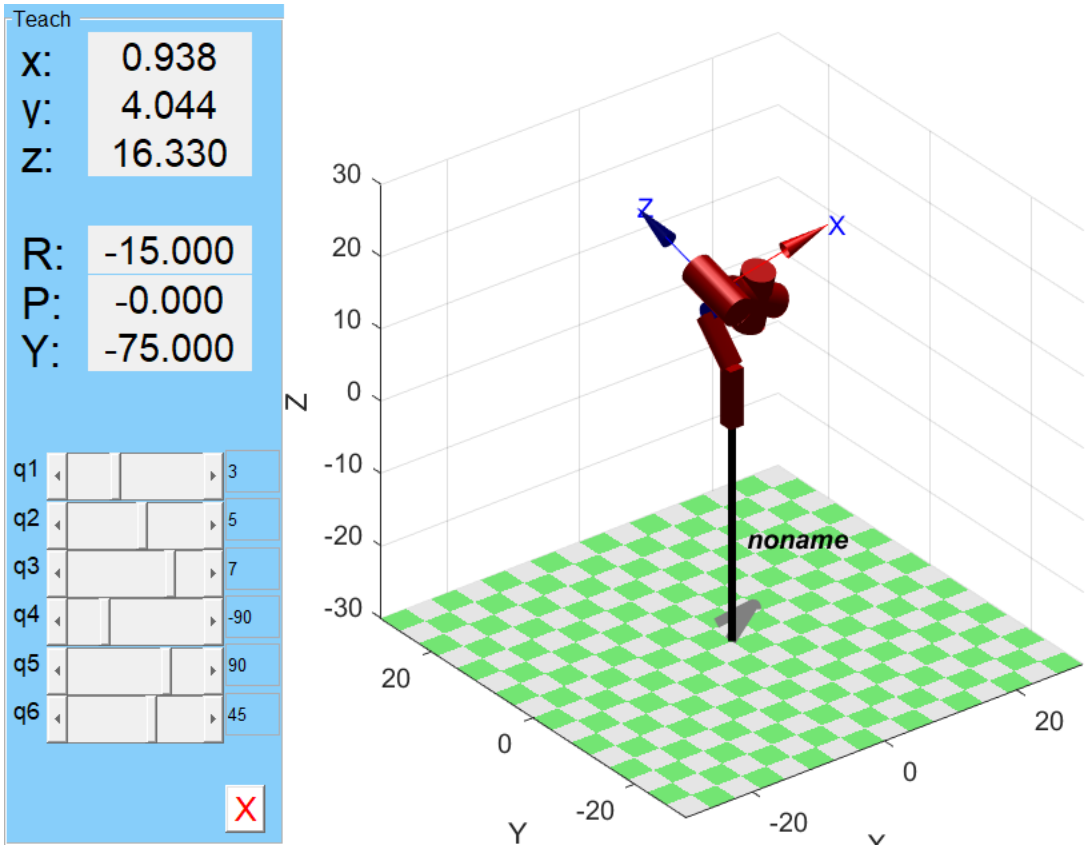
   12.2259
```

D:Plot the 6-dof manipulator including the teach function

```
   r.plot([3 5 7 -pi/2 pi/2 pi/4 ], 'workspace', [-30 30 -30 30 -30
30]);
```

```
    r.teach();
```

```
Warning: floor tiles too small, making them 20.000000 x bigger -
 change the size
or disable them
```



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